HARKIRAT SINGH

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PROFILE

Computer Engineering graduate proficient in Java, Python, database design, and REST API development. Strong background in building full-stack web applications and AI/ML integrations. Proven adaptability across technologies and domain frameworks, with solid grounding in Core CS fundamentals and team leadership.

EDUCATION

Thapar Institute of Engineering and Technology, Patiala — Patiala, Punjab
Bachelor of Engineering in Computer Engineering(CGPA: 7.23/10) — 2023 – 2027

WORK EXPERIENCE

AI Intern —XTRAGRAD Technologies Pvt. Ltd. — Aug 1, 2026 – Aug 30, 2026
Major Project: AI-Based Online Exam Proctoring & Abnormal Activity Detection
Tech Stack: Python, OpenCV, FastAPI, MongoDB, REST APIs, JWT, Git/GitHub

- Engineered an AI-driven Exam proctoring system that analyzes webcam and audio streams, leveraging **OpenCV** for face analysis, head-movement tracking, multi-person detection, and external-device identification during examinations.
- Architected a **risk-scoring and violation-tracking pipeline** with progressive warnings, live examiner monitoring, violation timelines, result analytics, and automated proctoring reports for high-risk session review.

Summer Project Intern —Experiential Learning Centre, TIET — June 2025 – July 2025

- Developed a passive, continuous authentication system for mobile banking using behavioral biometrics
- Implemented a Split Federated Learning architecture processing activations locally to keep raw sensor data on-device.
- Processed multi-modal mobile sensor streams including Accelerometer, Gyroscope logs, Keystroke dynamics, and Touch heatmaps to analyze user interaction patterns.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, OS, Computer Networks

Web Technology: NextJs, FastAPI, REST API Design, JWT, WebSockets

Generative AI: LangChain, Prompt Engineering, Vector Databases, RAG

Databases: MySQL, PostgreSQL, MongoDB, Redis

Tools & DevOps: Git, GitHub, GitHub Actions (CI/CD), Docker, Vercel, Render

Professional: Analytical Problem Solving, Communication, Collaboration, Leadership, Adaptability

PROJECTS

Certifications

KhataBox — *Inventory & Retail Management Platform* — Live Demo — Github
Python, TypeScript, NextJs, FastAPI, PostgreSQL, SQLAlchemy, Redis, Socket.IO, JWT Authentication

- Built a full-stack retail platform for inventory, POS billing, credit/Khata, orders, and multi-store management.
- Built real-time inventory tracking with QR scanning, low-stock alerts, automated receipts and live analytics.
- Developed dashboards with real-time sales, inventory, order, and profit analytics using Next.js & Shadcn UI.

SymptomScope AI — *AI/ML-Based Symptom Intelligence System* — Live Demo — Github
Python, TypeScript, NextJs, FastAPI, MongoDB, Clerk, LangChain, ChromaDB

- Built an AI symptom checker using Decision Tree, Random Forest & Naive Bayes for disease prediction, severity scoring.
- Engineered a Groq & LangChain RAG pipeline for structured, personalized recovery plans and medical guidance.
- Developed a goal-driven Groq AI agent with tool calling, context memory, and approval-based health actions.
- Built a RAG-based medical knowledge base using ChromaDB and TF-IDF embeddings for context-aware health responses.

IntelliMoney — *AI-Powered Financial Health Scoring Platform* — Live Demo — Github
Python, TypeScript, ReactJs, FastAPI, MongoDB, LangChain, OpenCV

- Engineered an AI-powered financial platform for expense tracking, budgeting, financial insights & health scoring.
- Developed a LangChain AI Copilot using Groq with tool-calling that answers finance queries and user-confirmed actions.
- Implemented TF-IDF & Logistic Regression for automated expense categorization, with OCR-based receipt extraction

LEADERSHIP & EXTRACURRICULAR

Joint Secretary– IETE Students Forum, TIET Patiala — May 2025 – May 2026

- Coordinated 10+ tech events and workshops, increasing club membership by 40%

Student Ambassador— LetsUpgrade EdTech Pvt. Ltd. — August 2024

- Led social outreach initiatives, boosting referral engagement by 20%.